

PROJECT OVERVIEW

Semantic Song Search Engine

Discovering songs by describing emotional themes and feelings, without knowing titles or artists.

Core Stack

Python · FastAPI

Embedding

Sentence Transformers

Vector Search

Milvus · HNSW

Dataset & Scope

57,650

Tracks before cleaning

Spotify Million Song Dataset (Kaggle)

Scope Limitation

Search targets lyrical emotional themes. The system does not measure instruments, tempo, or acoustic energy.

COLUMN	DESCRIPTION
artist	Artist name
song	Song title
link	Source link or path
text	Song lyrics content

System Architecture

OFFLINE: PREPARATION



ONLINE: QUERY HANDLING



Data Preprocessing

Preparing the song collection for embedding generation.



Clean Dataset

Remove empty lyrics and exact duplicates.



Preserve Meaning

Retain structural wording and vital negations.



Maintain Metadata

Keep vectors connected to artist and source.

Embedding Model

`sentence-transformers/all-MiniLM-L6-v2`

Converts text of varying lengths into uniform dense vectors.

- ✓ Pretrained model weights locked.
- ✓ All valid songs remain searchable.

Input Text

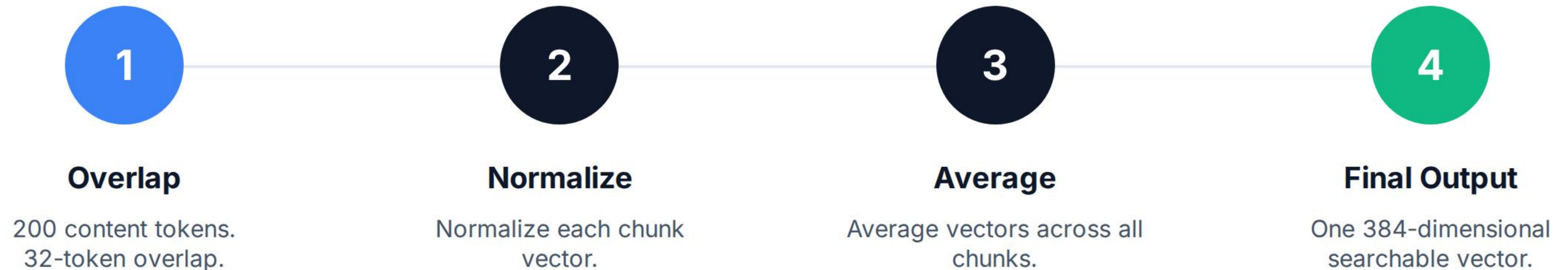


384 Dimensions

```
[0.124, -0.852, 0.441, ...  
-0.093, 0.776, 0.219, ...  
... 384 values total]
```

Chunking & Pooling

Addressing the 256-token model limit



SIMILARITY & RANKING

$$\textit{Cosine}(q, s) = q \cdot s$$

For unit-normalized vectors

Directional Match

Compares query direction with candidate vectors.

Raw Scoring

Display raw similarities, not probability percentages.

Milvus & HNSW Index

Storing embeddings and accelerating search with graph connections.

PROPOSED HNSW PARAMETERS

16

M

200

efConstruction

64

ef

10

top_k

FastAPI Web Interface

GET /
Search page rendering

POST /search
Query embedding & retrieval

GET /health
Search-engine readiness

Results Display

-  Rank and Similarity Score
-  Song Title and Artist
-  Short Lyrics Excerpt

Visible Disclaimer

"Results are based on lyrics and inferred themes, not audio analysis."

EVALUATION & TARGET METRICS

P@5

Precision at Top 5

nDCG@10

Ranking Quality

Lat

Query Latency

Connecting natural-language mood descriptions with song lyrics through embeddings and vector search.